

Comprehensive Data Analysis Report

Executive Summary

This report provides a comprehensive analysis of the provided dataset, which contains information about product sales across various regions. The objectives of this analysis are to understand the key statistics, correlations, and data quality issues in the dataset, and to provide recommendations for further analysis and potential improvements.

Key Findings

Descriptive Statistics

• **ProductionCost:**

• Count: 2000

• Mean: 499.97

• Standard Deviation: 98.84

• Minimum: 129.12

• 25%: 434.45

• Median: 500.60

• 75%: 566.59

• Maximum: 865.77

• **MarketingSpend:**

• Count: 2000

• Mean: 201.27

• Standard Deviation: 49.96

• Minimum: 46.78
• 25%: 167.00
• Median: 201.05
• 75%: 235.15
• Maximum: 379.97

• **SeasonalDemandIndex:**

• Count: 2000
• Mean: 1.48
• Standard Deviation: 0.50
• Minimum: -0.53
• 25%: 1.15
• Median: 1.49
• 75%: 1.83
• Maximum: 3.05

• **CompetitorPrice:**

• Count: 2000
• Mean: 299.59
• Standard Deviation: 81.52
• Minimum: 46.02
• 25%: 241.27
• Median: 301.35
• 75%: 355.09
• Maximum: 552.44

• **CustomerRating:**

• Count: 2000
• Mean: 3.99
• Standard Deviation: 0.50
• Minimum: 2.45

• 25%: 3.66

• Median: 3.99

• 75%: 4.33

• Maximum: 5.55

• **EconomicIndex:**

• Count: 2000

• Mean: 100.22

• Standard Deviation: 19.18

• Minimum: 36.00

• 25%: 87.30

• Median: 99.95

• 75%: 112.60

• Maximum: 164.32

• **StoreCount:**

• Count: 2000

• Mean: 48.67

• Standard Deviation: 28.94

• Minimum: 1.00

• Median: 49.00

• 75%: 74.00

• Maximum: 99.00

• **SalesRevenue:**

• Count: 2000

• Mean: 2072.86

• Standard Deviation: 346.01

• Minimum: 1094.52

• 25%: 1808.11

• Median: 2068.17

• 75%: 2341.22

• Maximum: 3115.11

Correlation Analysis

- **ProductionCost vs. SalesRevenue:** This correlation is moderately strong, with a coefficient of 0.34.
- **StoreCount vs. SalesRevenue:** This correlation is very strong, with a coefficient of 0.82.

Data Quality Assessment

Missing Values

- There are no missing values in the dataset.

Data Types

- There are 11 columns in the dataset, with the following data types:

• ProductCategory: object

• Region: object

• CustomerSegment: object

• IsPromotionApplied: object

• ProductionCost: float64

• MarketingSpend: float64

• SeasonalDemandIndex: float64

• CompetitorPrice: float64

• CustomerRating: float64

• EconomicIndex: float64

• StoreCount: int64

• SalesRevenue: float64

Data Distribution

• The distribution of the data is as follows:

• ProductionCost: Right-skewed

• MarketingSpend: Left-skewed

• SeasonalDemandIndex: Normal

• CompetitorPrice: Right-skewed

• CustomerRating: Normal

• EconomicIndex: Normal

• StoreCount: Skewed to the right

• SalesRevenue: Right-skewed

• There are 2 outliers in the dataset:

• ProductionCost: 865.77

• SalesRevenue: 3115.11

Recommendations

1. **Analyze the impact of ProductionCost on SalesRevenue:** Further analysis is needed to understand the relationship between ProductionCost and SalesRevenue, and to determine if there are any trends or patterns that can be used to improve sales.

2. **Investigate factors influencing StoreCount:** The strong correlation between StoreCount and SalesRevenue suggests that factors influencing StoreCount may also impact sales. Further investigation is needed to understand these factors and how they can be utilized to increase sales.

Limitations and Future Work

1. **Limited sample size:** Although the dataset contains 2000 rows, it is possible that the sample size is limited and may not be representative of the larger population.

2. **Lack of contextual information:** The dataset does not provide any contextual information about the product or sales environment, which makes it difficult to interpret the results.

Conclusion

In conclusion, this analysis provides a comprehensive understanding of the dataset and highlights key statistics, correlations, and data quality issues. The recommendations provided are based on the findings and aim to improve the understanding and utilization of the dataset. Future work should focus on addressing the limitations and investigating potential factors influencing StoreCount and ProductionCost.